TRUTH VS DECEPTION

The AI Courtroom: Putting Fake News on Trial

PRESENTED BY: THE LAN-guage MODELS

Laxmi | Alan | Nicole

The Case Against Misinformation

Fake news spreads 6 times faster than truth. It manipulates public opinion and erodes trust in global institutions. Our project builds a digital defense—a machine learning jury designed to verify every claim.

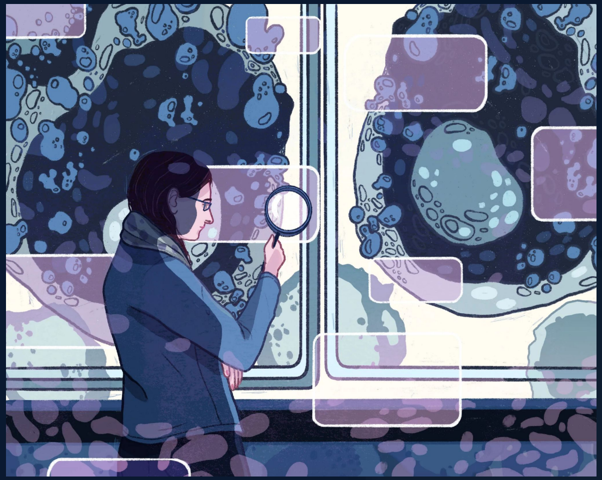
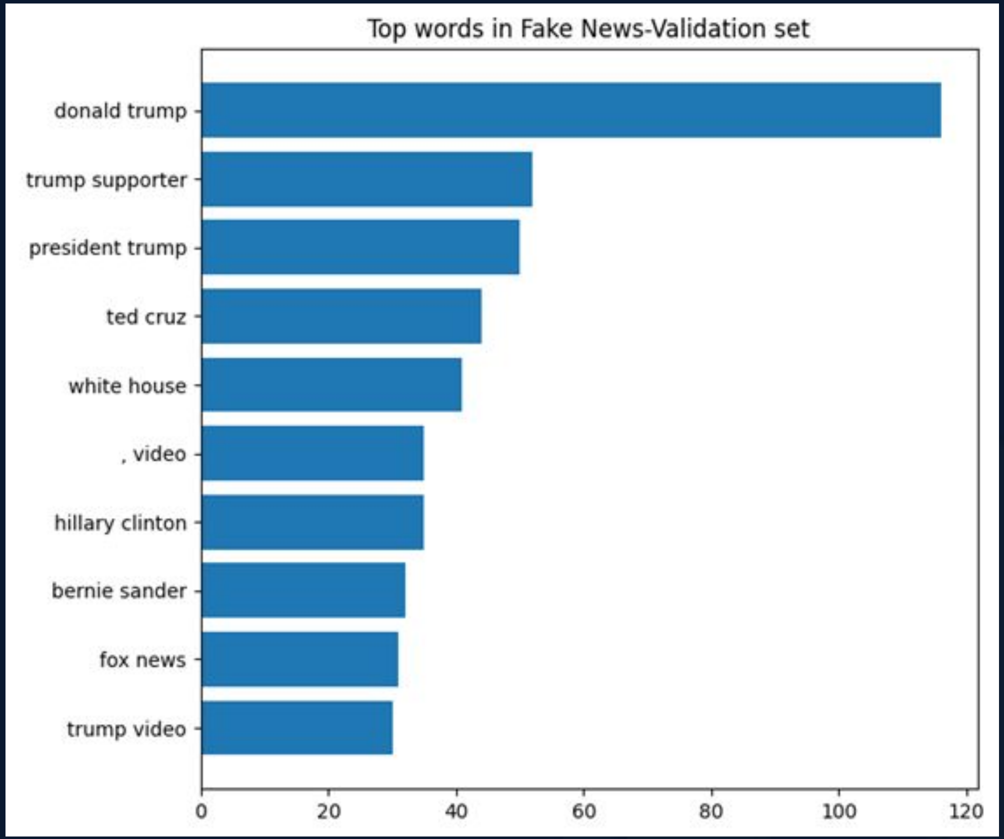


EXHIBIT A: THE EVIDENCE ARCHIVE

Our evidence vault contains over **34,000 headlines previously audited, and ~10 000 to** classify. Every file is a witness to the patterns of deceptive writing.

- **Corpus:** Real & Fake News Archive
- **Analysis:** NLP Classification
- **Integrity:** 50/50 Dataset Balance



FORENSIC TEXT CLEANING

Noise Removal

Stripping away punctuation and special characters that distract from the core linguistic message.

Evidence Filtering

Removing 'Stopwords'—common linguistic filler that carries no investigative weight in deception detection.

Lemmatizing

Reducing diverse words to their root forms to ensure the AI jury recognizes deceptive patterns across different tenses.

UNCOVERING LINGUISTIC CUES

TF-IDF

FORENSIC WEIGHTING

We transform text into **Numerical Evidence**. Using TF-IDF and Bag of Words, we weigh the significance of every phrase, identifying the unique "word signatures" of fake news headlines.

THE AI JURY PANEL

Juror Nicole
The Reliable Baseline Witness

SVM Classifier

Juror Alan
The Heavyweight Judge

DistilBERT Transformer

CHIEF JUSTICE

Juror Laxmi
The Semantic Authority

BERT Tiny Chief Justice

THE DELIBERATION:

We chose a diverse panel. Nicole represents classical statistical learning, while Alan and Laxmi represent the modern 'transformer' era of NLP, which understands the deeper context and semantics of the news articles.

JUROR: NICOLE

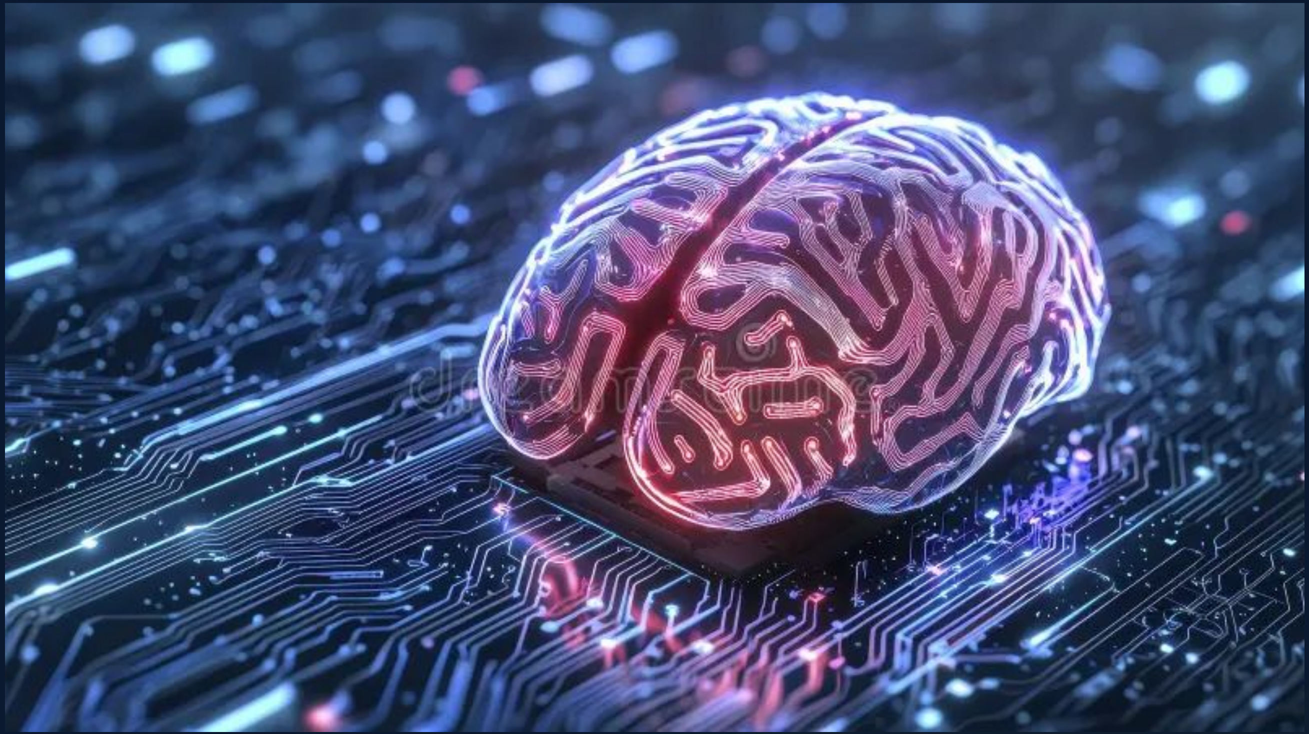
The Reliable Witness

Nicole's model utilizes a Support Vector Machine (SVM) with TF-IDF Unigrams and Bigrams to improve resolution of the evidence context. Evaluation metrics relied on the least amount of False Negatives.

93.6%

ACCURACY VERDICT

JUROR: ALAN



The Heavyweight Judge

Alan's model uses DistilBERT, a lightweight transformer architecture that understands contextual relationships and semantic meaning within text. Despite being smaller and faster than traditional BERT models, it preserves strong language understanding capabilities while significantly improving computational efficiency.

96.18%

ACCURACY VERDICT

ALAN'S PERFORMANCE:

Alan's DistilBERT model moves beyond simple word counts. It looks at how words relate to each other bidirectionally. This context-aware approach raises the accuracy to 96.18%, identifying subtle deceptive narratives.

CHIEF JUSTICE VERDICT

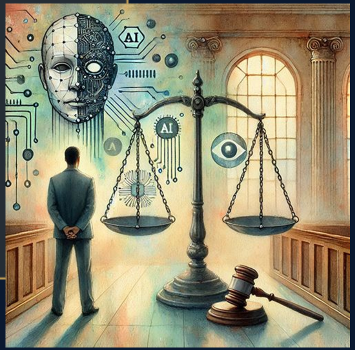
JUROR: LAXMI

The Semantic Authority

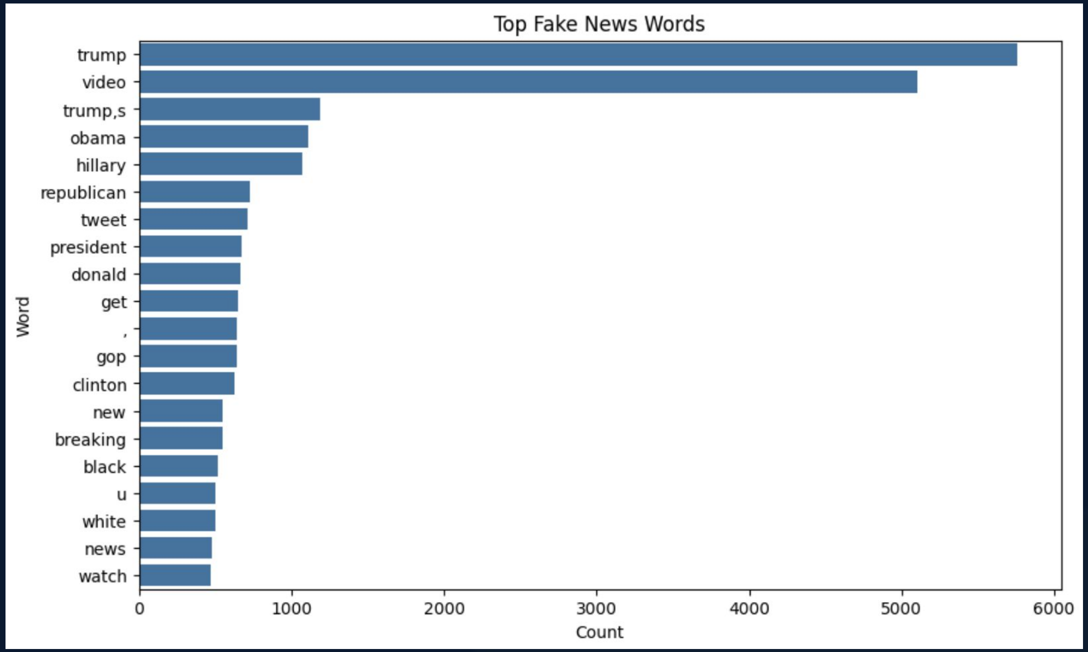
Laxmi's BERT Tiny model stands as our Chief Justice. Despite its lightweight architecture, it outperformed all other models by identifying the deepest semantic patterns within the evidence.

96.7%

CHIEF ACCURACY



The suspects identified



LAXMI'S PERFORMANCE:

Chief Justice Laxmi achieved our top accuracy of 96.7% using BERT Tiny. This proves that a well-tuned transformer model, even a small one, is the ultimate judge when it comes to understanding deceptive semantics.

It was of sure help to identify the fake news words

JURY DELIBERATION RESULTS

Across all metrics—Accuracy, Precision, and F1-Score—our AI jury demonstrated exceptional judgment.

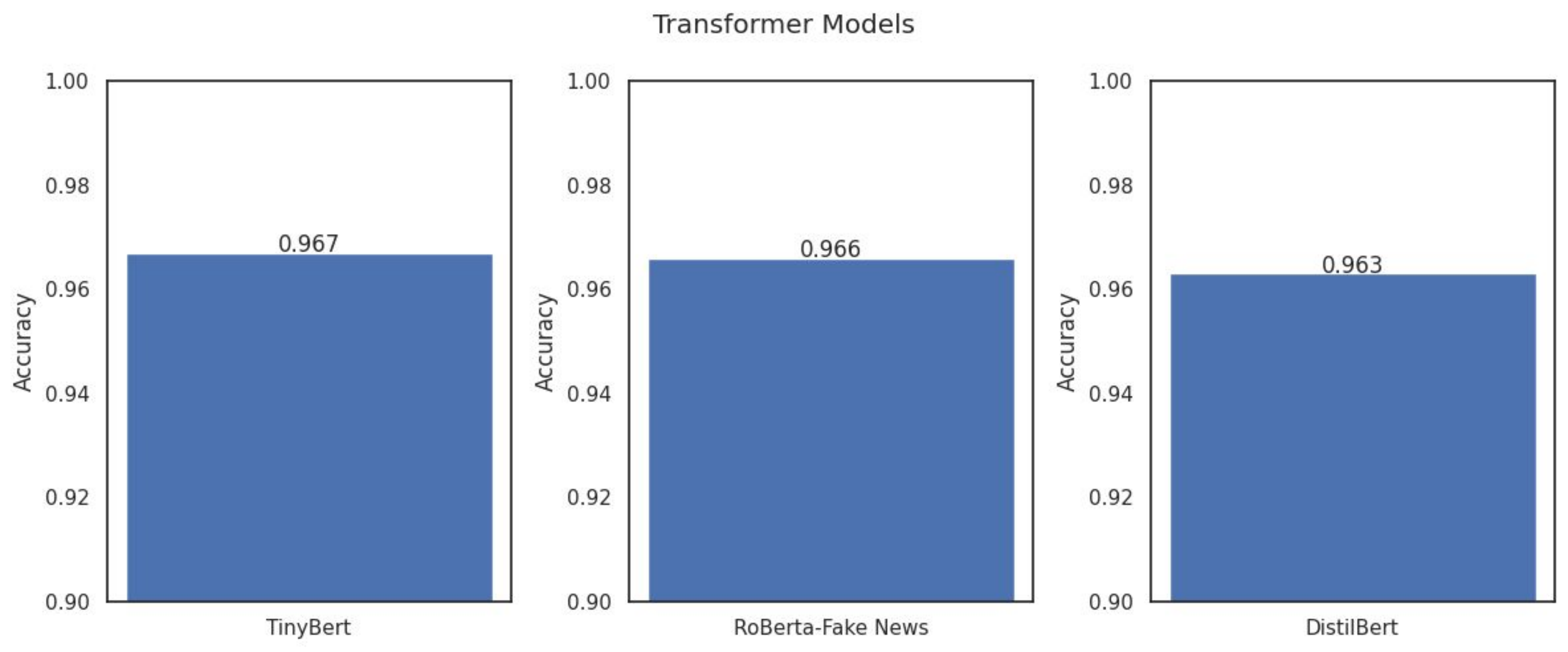
THE FINAL TALLY:

The results are undeniable. While Nicole's baseline is excellent, the Transformer models (Alan and Laxmi) provide the extra precision required to win the case against sophisticated fake news campaigns.

Transformed Jurors

Our transformed jurors represent the next generation of NLP: transformer models capable of analyzing context, semantics, and deeper linguistic patterns beyond simple keyword detection.

Juror Laxmi | **Juror Nicole** | **Juror Alan**



CROSS-EXAMINATION

- **Sarcasm & Satire:** Models sometimes mistake heavy satire for outright deception, lacking the "human" sense of irony.
- **Contextual Drift:** News from 2024 might contain language patterns not found in our 2017 training evidence.
- **Noisy Data:** Extremely short headlines provide too little evidence for the jury to reach a confident verdict.
- **Overfitting:** The risk of the jury "memorizing" specific dates rather than learning linguistic patterns.

THE AFTER-ACTION:

Every courtroom has its hurdles. We learned that the quality of our forensic evidence prep—the cleaning—is what truly enables the judges to reach these high accuracy scores.

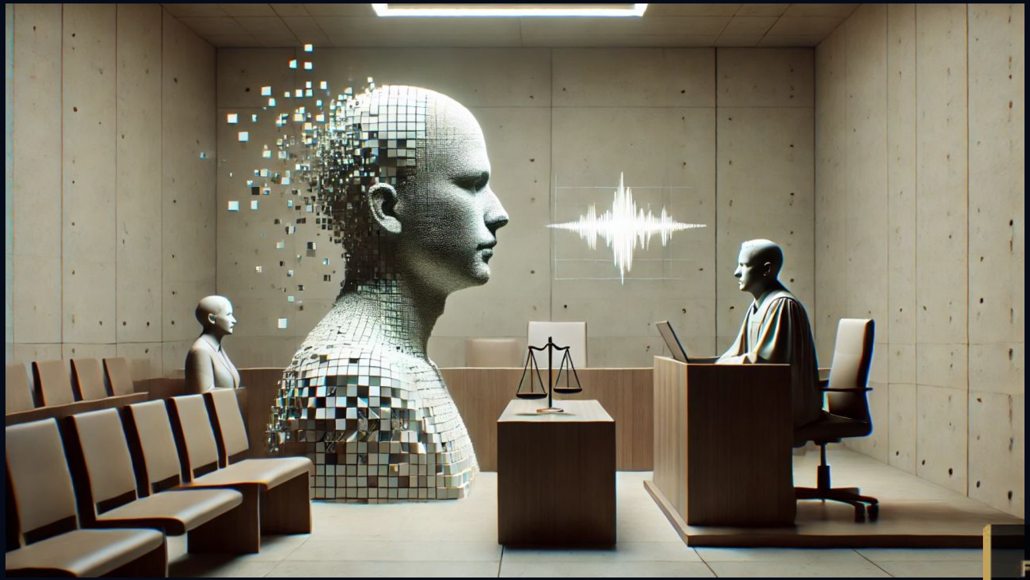
LESSONS

Lessons

- Preprocessing is 70% of the battle.
- Model size doesn't always equal accuracy.
- Context is king in truth detection.

FINAL JUDGMENT

Case Closed. Thank You.



FINAL ADJOURNMENT:

We have proven that while deception is evolving, truth is detectable. The case is adjourned. We are now open for cross-examination—do you have any questions for the jury?

THE LAN-GUAGE MODELS

NICOLE · ALAN · LAXMI